In [53]:
```
import seaborn as sns
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

In [54]:
```
df=sns.load_dataset('titanic')
df.head()
```

Out[54]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alon
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	Fals
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	Fals
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	Tru
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	Fals
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	Tru

In [55]:
```
print(df.head())
```

```
   survived  pclass     sex   age  sibsp  parch     fare embarked  class  \
0         0       3    male  22.0      1      0   7.2500        S  Third
1         1       1  female  38.0      1      0  71.2833        C  First
2         1       3  female  26.0      0      0   7.9250        S  Third
3         1       1  female  35.0      1      0  53.1000        S  First
4         0       3    male  35.0      0      0   8.0500        S  Third

     who  adult_male deck  embark_town alive  alone
0    man        True  NaN  Southampton    no  False
1  woman       False    C    Cherbourg   yes  False
2  woman       False  NaN  Southampton   yes   True
3  woman       False    C  Southampton   yes  False
4    man        True  NaN  Southampton    no   True
```

In [56]:
```
df.tail()
```

Out[56]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	al
886	0	2	male	27.0	0	0	13.00	S	Second	man	True	NaN	Southampton	no	T
887	1	1	female	19.0	0	0	30.00	S	First	woman	False	B	Southampton	yes	T
888	0	3	female	NaN	1	2	23.45	S	Third	woman	False	NaN	Southampton	no	F
889	1	1	male	26.0	0	0	30.00	C	First	man	True	C	Cherbourg	yes	T
890	0	3	male	32.0	0	0	7.75	Q	Third	man	True	NaN	Queenstown	no	T

In [57]:
```
df.info()
```

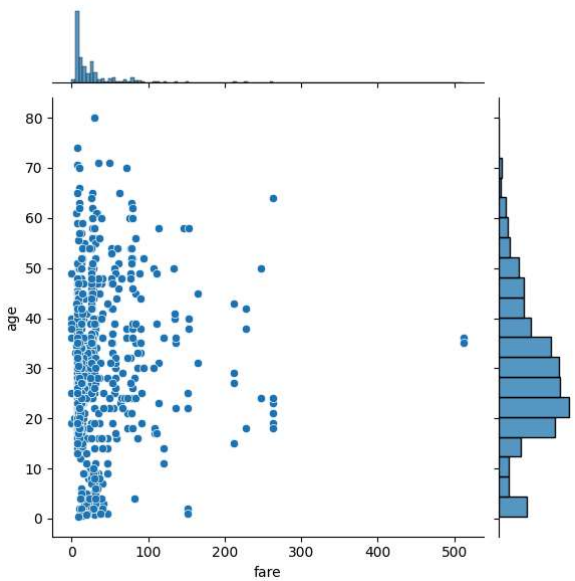
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     891 non-null    int64
 1   pclass       891 non-null    int64
 2   sex          891 non-null    object
 3   age          714 non-null    float64
 4   sibsp        891 non-null    int64
 5   parch        891 non-null    int64
 6   fare         891 non-null    float64
 7   embarked     889 non-null    object
 8   class        891 non-null    category
 9   who          891 non-null    object
 10  adult_male   891 non-null    bool
 11  deck         203 non-null    category
 12  embark_town  889 non-null    object
 13  alive        891 non-null    object
 14  alone        891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
```

In [58]:
```
df.shape
```

Out[58]: (891, 15)

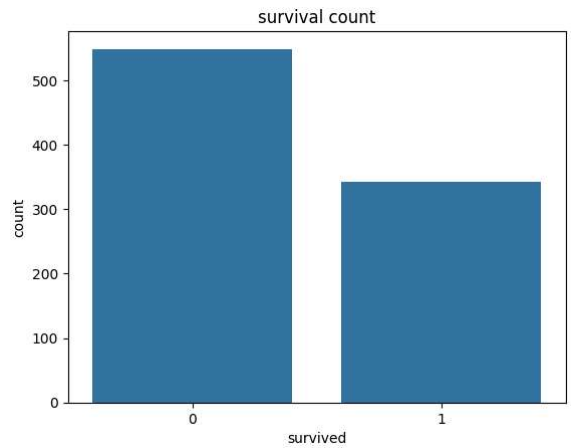
In [59]:
```
sns.jointplot(x='fare',y='age',data=df)
```

Out[59]: <seaborn.axisgrid.JointGrid at 0x72f608a69420>



In [60]:
```
#to visuloaize patteren in seaborn
sns.countplot(x='survived',data=df)
```

```
plt.title('survival count')
plt.show()
```

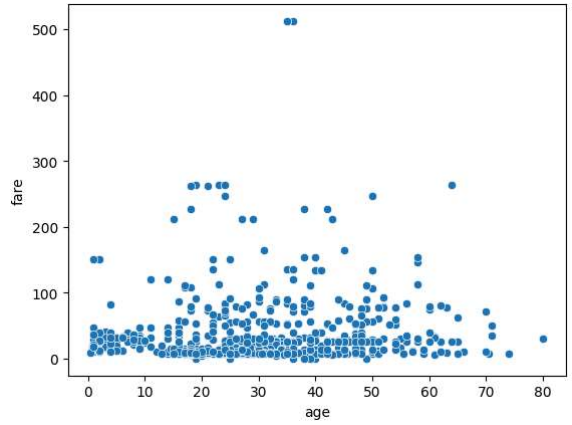


In [61]:
```
df.isnull().sum()
```

Out[61]:
```
survived         0
pclass           0
sex              0
age            177
sibsp            0
parch            0
fare             0
embarked         2
class            0
who              0
adult_male       0
deck           688
embark_town      2
alive            0
alone            0
dtype: int64
```

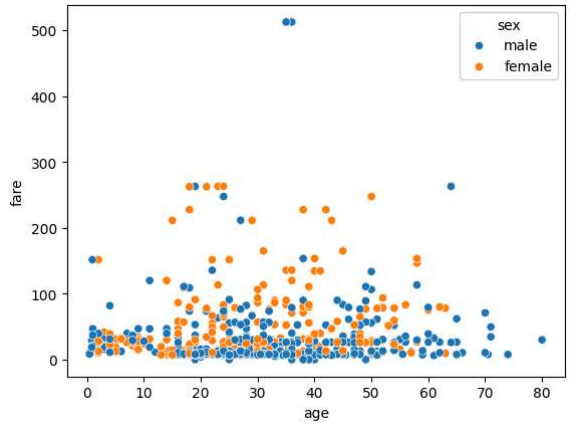
In [62]:
```
#scatterplot used to check the outlier of the column fare with respect to coloumn age
sns.scatterplot(data=df,x='age',y='fare')
```

Out[62]:
```
<Axes: xlabel='age', ylabel='fare'>
```



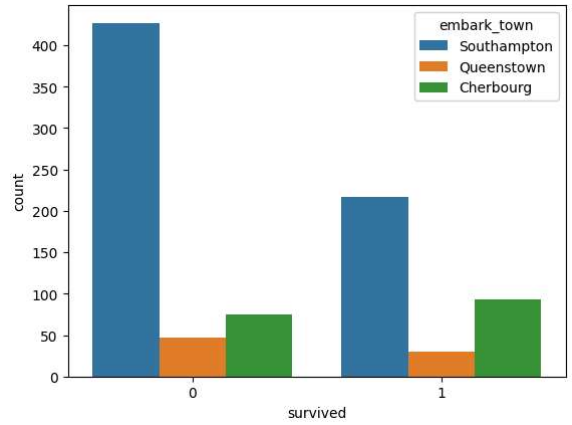
In [63]:
```
#categories accoding the the sex and hue is used to it
sns.scatterplot(data=df,x='age',y='fare',hue='sex')
```

Out[63]:
```
<Axes: xlabel='age', ylabel='fare'>
```



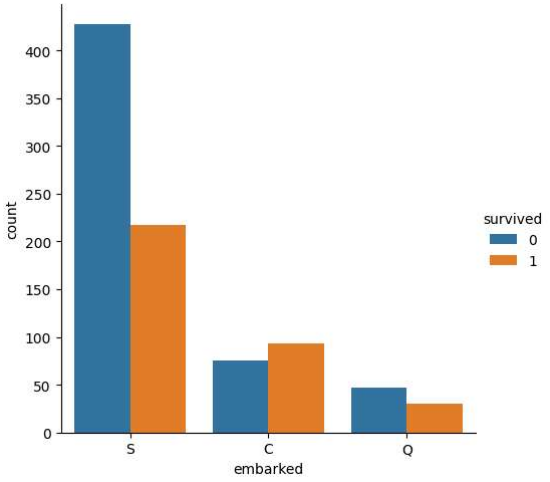
In [64]:
```
#countplot -to determine number of ppl survived  and not survived
sns.countplot(data=df,x='survived',hue='embark_town')
```

Out[64]:
```
<Axes: xlabel='survived', ylabel='count'>
```



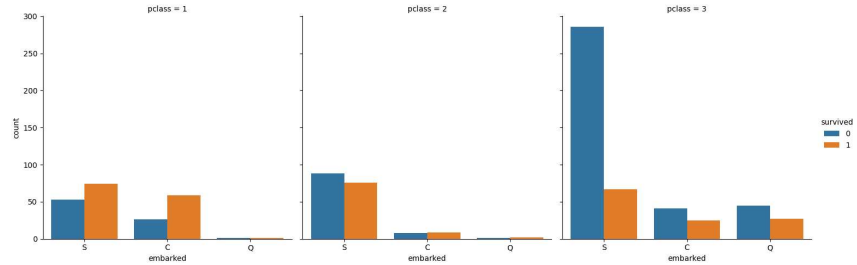
In [65]:
```
sns.catplot(data=df,x='embarked',hue='survived',kind='count')
```

Out[65]:
```
<seaborn.axisgrid.FacetGrid at 0x72f6087c0b50>
```



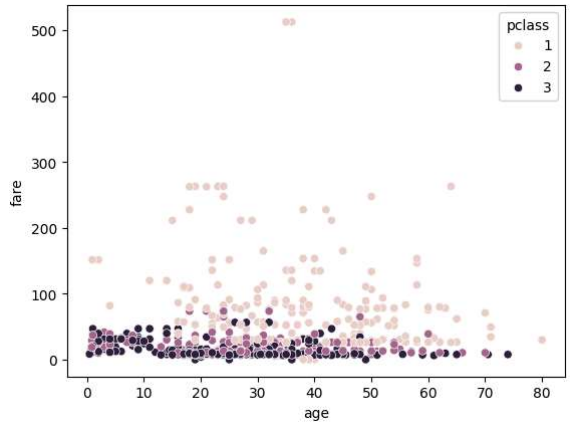
In [66]:
```
sns.catplot(x='embarked',hue='survived',data=df,kind='count',col='pclass')
```

Out[66]: <seaborn.axisgrid.FacetGrid at 0x72f608ca84c0>



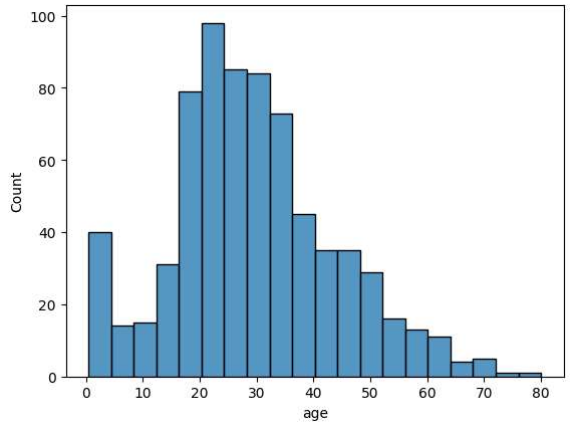
In [67]:
```
sns.scatterplot(data=df,x='age',y='fare',hue='pclass')
```

Out[67]: <Axes: xlabel='age', ylabel='fare'>



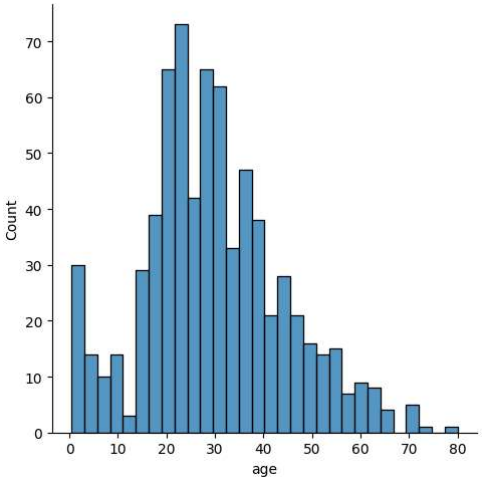
In [68]:
```
sns.histplot(data=df,x='age')
```

Out[68]: <Axes: xlabel='age', ylabel='Count'>



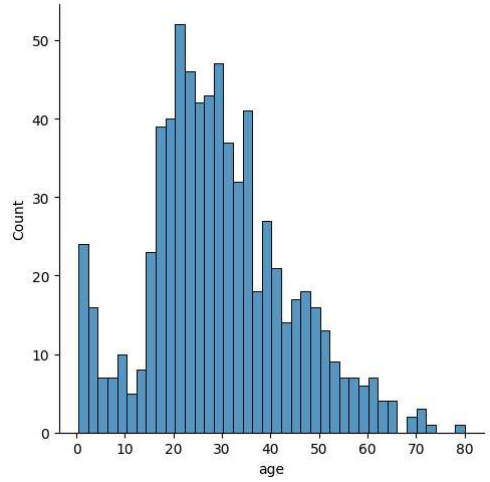
In [69]:
```
sns.displot(data=df,x='age',bins=30)
```

Out[69]: <seaborn.axisgrid.FacetGrid at 0x72f6082b4eb0>



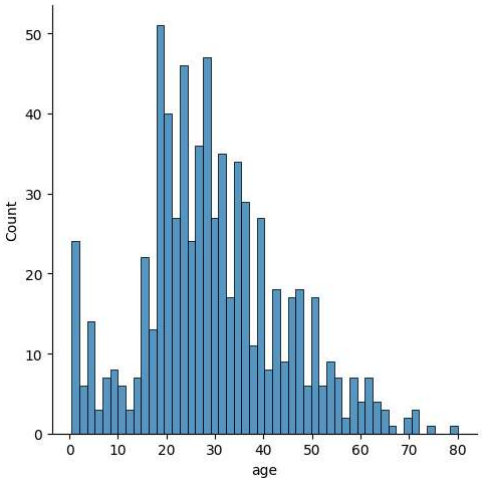

```
In [70]: sns.displot(data=df,x='age',bins=40)
```

```
Out[70]: <seaborn.axisgrid.FacetGrid at 0x72f608b279d0>
```



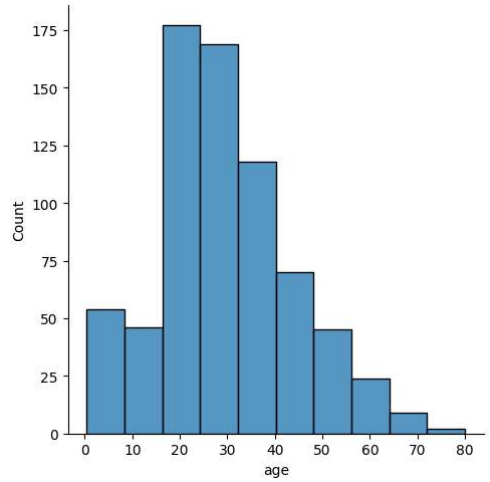

```
In [71]: sns.displot(data=df,x='age',bins=50)
```

```
Out[71]: <seaborn.axisgrid.FacetGrid at 0x72f6089ecbb0>
```



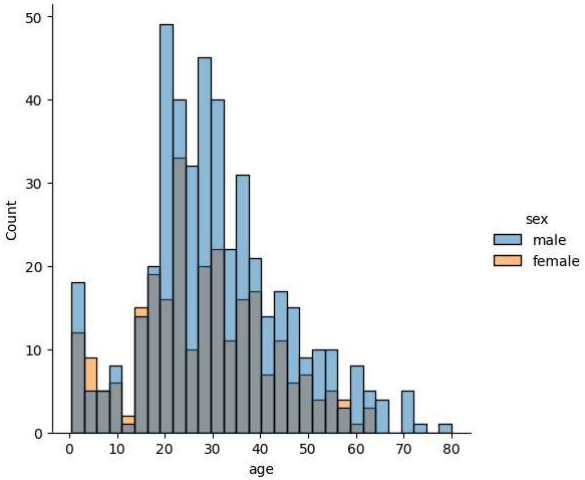

```
In [72]: sns.displot(data=df,x='age',bins=10)
```

```
Out[72]: <seaborn.axisgrid.FacetGrid at 0x72f609288e20>
```



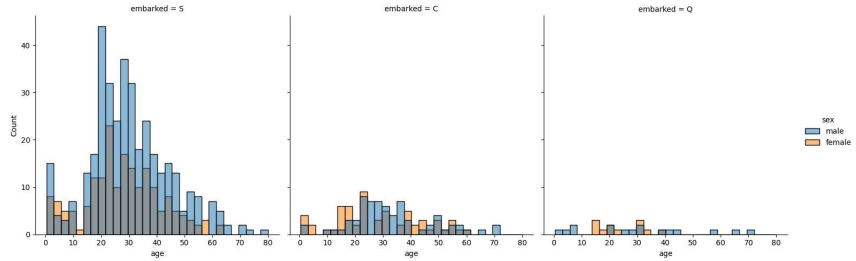

```
In [73]: sns.displot(data=df,x='age',bins=30,hue='sex')
```

```
Out[73]: <seaborn.axisgrid.FacetGrid at 0x72f6082b4700>
```



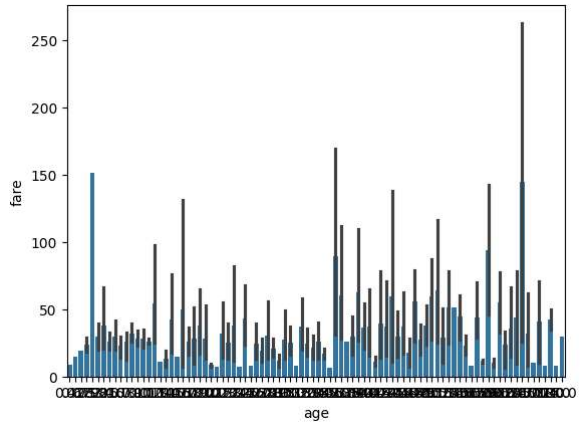
In [74]:
```
sns.displot(data=df,x='age',bins=30,hue='sex',col='embarked')
```

Out[74]:
```
<seaborn.axisgrid.FacetGrid at 0x72f6087661a0>
```



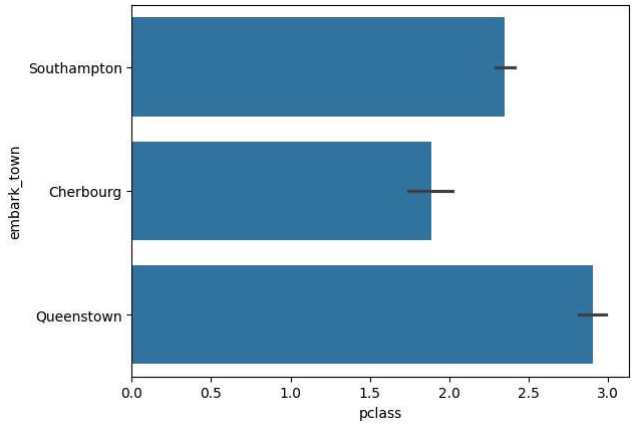
In [75]:
```
sns.barplot(x='age',y='fare',data=df)
```

Out[75]:
```
<Axes: xlabel='age', ylabel='fare'>
```



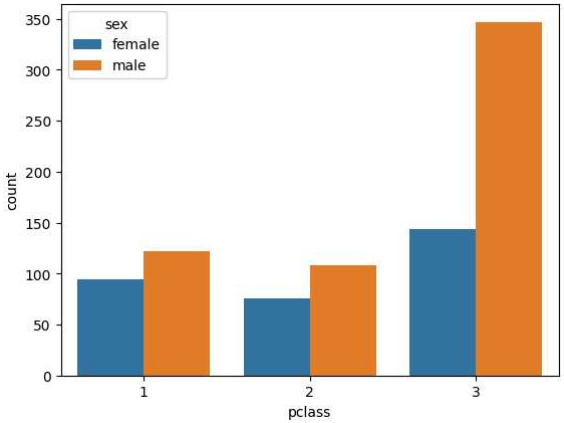
In [76]:
```
sns.barplot(data=df,x='pclass',y='embark_town',orient='h')
```

Out[76]:
```
<Axes: xlabel='pclass', ylabel='embark_town'>
```



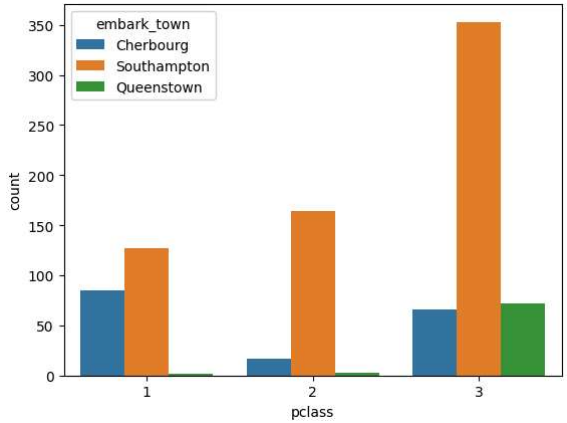
In [77]:
```
sns.countplot(data=df,x='pclass',hue='sex')
```

Out[77]:
```
<Axes: xlabel='pclass', ylabel='count'>
```



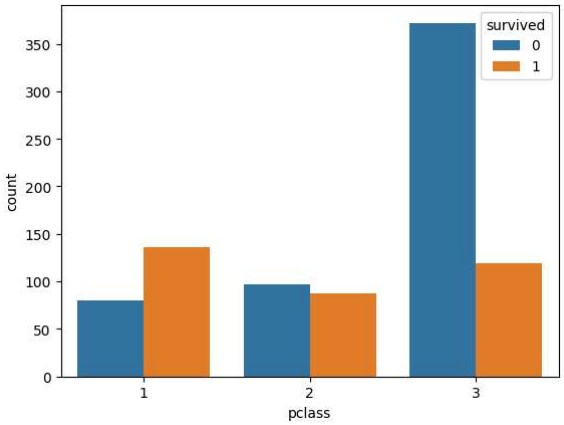
In [78]:
```
sns.countplot(data=df,x='pclass',hue='embark_town')
```

Out[78]: <Axes: xlabel='pclass', ylabel='count'>



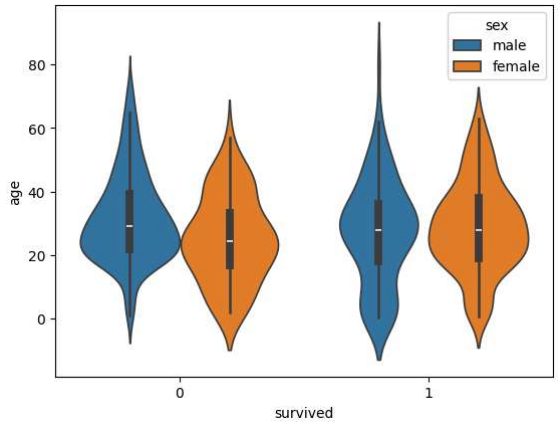
In [79]:
```
sns.countplot(data=df,x='pclass',hue='survived')
```

Out[79]: <Axes: xlabel='pclass', ylabel='count'>



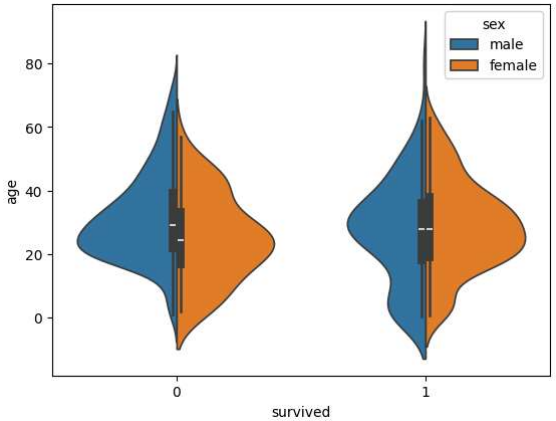
In [80]:
```
sns.violinplot(data=df,x='survived',y='age',hue='sex')
```

Out[80]: <Axes: xlabel='survived', ylabel='age'>



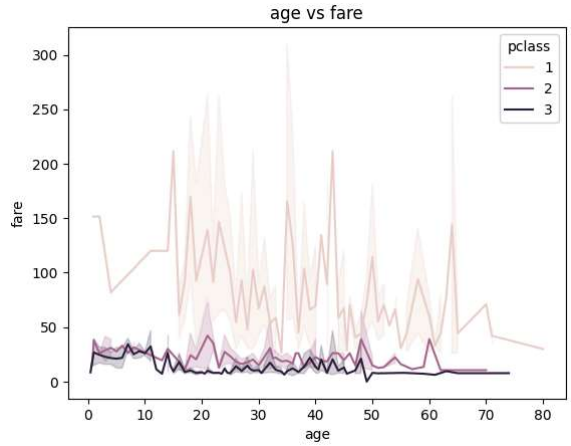
In [81]:
```
sns.violinplot(data=df,x='survived',y='age',hue='sex',split=2)
```

Out[81]: <Axes: xlabel='survived', ylabel='age'>



In [82]:
```
#timeseries and dat analysis--trend analysis
sns.lineplot(data=df,x='age',y='fare',hue='pclass')
plt.title('age vs fare')
```

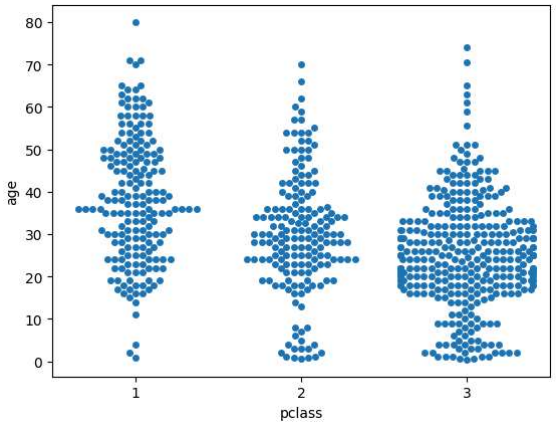
Out[82]: Text(0.5, 1.0, 'age vs fare')



In [83]:
```
sns.swarmplot(data=df,x='pclass',y='age')
```

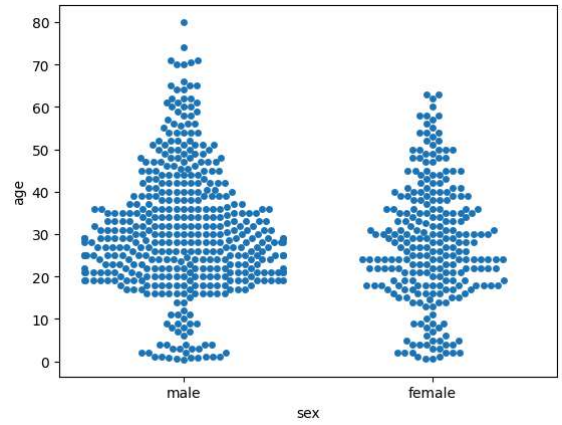
Out[83]: <Axes: xlabel='pclass', ylabel='age'>

```
/home/student/.local/lib/python3.10/site-packages/seaborn/categorical.py:3399: UserWarning: 15.2% of the points canno
t be placed; you may want to decrease the size of the markers or use stripplot.
  warnings.warn(msg, UserWarning)
```

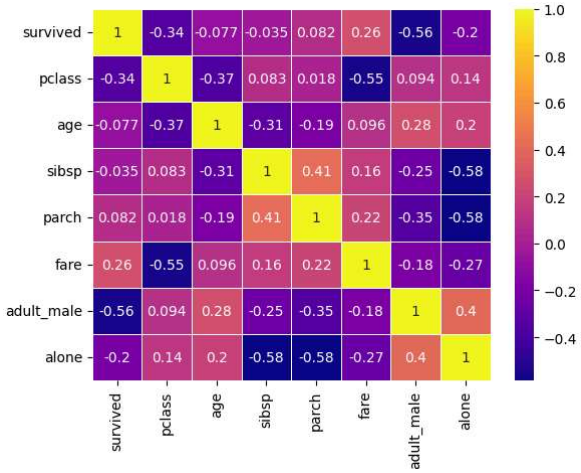


In [84]:
```
sns.swarmplot(data=df,x='sex',y='age')
```

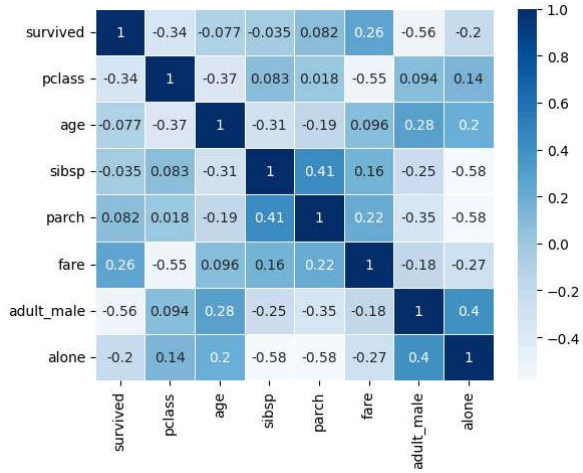
Out[84]: <Axes: xlabel='sex', ylabel='age'>



In [86]:
```
corr=df.corr(numeric_only=True)
sns.heatmap(corr,annot=True,cmap='plasma',linewidths=0.5)
plt.show()
```



In [88]:
```
corr=df.corr(numeric_only=True)
sns.heatmap(corr,annot=True,cmap='Blues',linewidths=0.5)
plt.show()
```



In [ ]: